SAMEEKSHA MAHESH

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SUMMARY

Master's student in Computer Science at UC Davis with hands-on experience building production microservices and full-stack platforms. As a Software Development Engineer, redesigned Java Spring Boot services to cut API response times by 20% and delivered an OCR-based bank identification system boosting first-pass accuracy by 35%. Skilled in Python, Java, AWS, Docker and end-to-end application development in agile, cross-functional teams.

EDUCATION

University of California, Davis **Mar 2027**
Master of Science in Computer Science (GPA: 4.0/4.0)
- **Coursework:** Distributed Database Systems, Analysis of Software Artifacts

Dayanand Sagar College of Engineering **2024**
Bachelor of Engineering in Computer Science (GPA: 3.97/4.0)
- **Coursework:** Data Structures and Algorithms, Operating System, Big Data, Cloud Computing, Object Oriented Programming

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, JavaScript, Go, Perl
- **Frameworks & Libraries:** Spring Boot, Flask, Node.js, Rest APIs, Elasticsearch, Microservices
- **Developer Tools:** Git, Docker, Kubernetes, Jenkins, GitHub Actions, Postman, Linux
- **Databases:** MySQL, PostgreSQL, MongoDB, Oracle
- **Cloud / DevOps:** AWS (EC2, Lambda, EBS, S3), CI/CD pipeline

EXPERIENCE

Perfios Software Solutions Pvt Ltd | *Software Development Engineer* **Apr 2024 - Sep 2025**
- Redesigned **RESTful** microservices in Java and Spring Boot using efficient data structures and object-oriented design principles, optimizing SQL queries and indexing to cut API response time by **20%** and improve system maintainability.
- Created a **full-stack** invoicing platform with Node.js, PostgreSQL, and role-based access control (RBAC), conducting experiment design to validate reporting metrics and partnering cross-functionally to accelerate data-driven decision making.
- Delivered an OCR-based bank identification system for international banks using Python and computer vision libraries, improving first-attempt recognition accuracy by **35%**.
- Containerized and deployed services using Docker and Kubernetes on AWS EC2 and Lambda, and established a CI/CD pipeline via GitHub Actions and Jenkins, improving deployment reliability and reducing release cycles.

Brane Enterprises Pvt. Ltd | *Automation Intern* **Jun 2022 - Jan 2023**
- Automated **API workflows** using Python scripts and Postman and integrated them into a CI/CD pipeline with Jenkins, applying experiment design and data analysis to increase reliability and reduce manual testing effort by **15%**.
- Optimized client-facing automation tools in Node JS, working autonomously under **Agile timelines** and using Git for version control to improve platform performance.

PROJECTS

Intelligent Traffic Violation Detection System **Mar 2024 - Jun 2024**
- Created a real-time traffic violation detection system for helmet non-compliance, signal violations, and mobile phone usage using deep learning and computer vision (YOLOv3/v5/v8 + SORT).
- Integrated Tesseract OCR for automatic license plate recognition and built a Streamlit web interface enabling video upload, real-time violation visualization, and MySQL-based record storage.
- Achieved 73% mAP50 on validation datasets, enabling accurate and efficient monitoring of traffic violations.

Secure Healthcare Portal **Oct 2023 - Feb 2024**
- Engineered a secure, cloud-based telemedicine platform for encrypted storage and sharing of medical records between patients and doctors, ensuring confidentiality, integrity, and reliable access.
- Implemented RSA-2048 encryption, secure key management, and two-factor authentication via Flask REST APIs, achieving 100% data confidentiality in security testing across 500+ transactions.

Fake News Detection System **Feb 2023 - May 2023**
- Built a fake news detection system using NLP and ensemble ML models (Logistic Regression, SVM, Random Forest) with a text preprocessing pipeline (tokenization, TF-IDF vectorization), achieving 96% classification accuracy.
- Deployed an interactive Flask web interface for real-time article submission and credibility scoring with confidence metrics.